NOVA SCHOOL OF SCIENCE & TECHNOLOGY

Nucleo de AeroEspacial

ASTRO

Electronics Report

Author:

Lorem Ipsum (31416) lorem.ipsum@campus.fct.unl.pt
Lorem Ipsum (31416) lorem.ipsum@campus.fct.unl.pt

2025/2026 – 1st Semester, DEEC

Contents

List of Figures

1 Introduction

This document presents the Design Review of the avionics and simulation systems developed for the sounding rocket project by the Astro team. The objective of this review is to describe the design choices, implementation details, and validation methods adopted for the onboard electronics, communication systems, ground station, and trajectory simulations, while demonstrating compliance with mission and competition requirements.

The avionics subsystem is responsible for sensor data acquisition, flight state estimation, telemetry transmission, recovery system activation, and overall mission safety. Reliability, redundancy, and robustness under high-dynamic flight conditions were key drivers throughout the design process. Component selection and system architecture were therefore guided by prior flight experience, established aerospace practices, and simulation-based validation.

In parallel, detailed trajectory simulations were conducted to predict flight performance, assess aerodynamic behaviour, and support design decisions related to propulsion, stability, and recovery systems. These simulations serve both as a verification tool during development and as a means of reducing risk ahead of flight operations.

This report is structured as follows: Section 2 describes the sensors and analog circuitry; Section 3 presents the selected microcontroller; Section 4 details the communication system; Section 5 outlines the ground station architecture; and Section 6 discusses the simulation framework and validation methodology. Supporting material is provided in the annexes.

2 Electronic Components

2.1 Sensors and Analogue Circuit

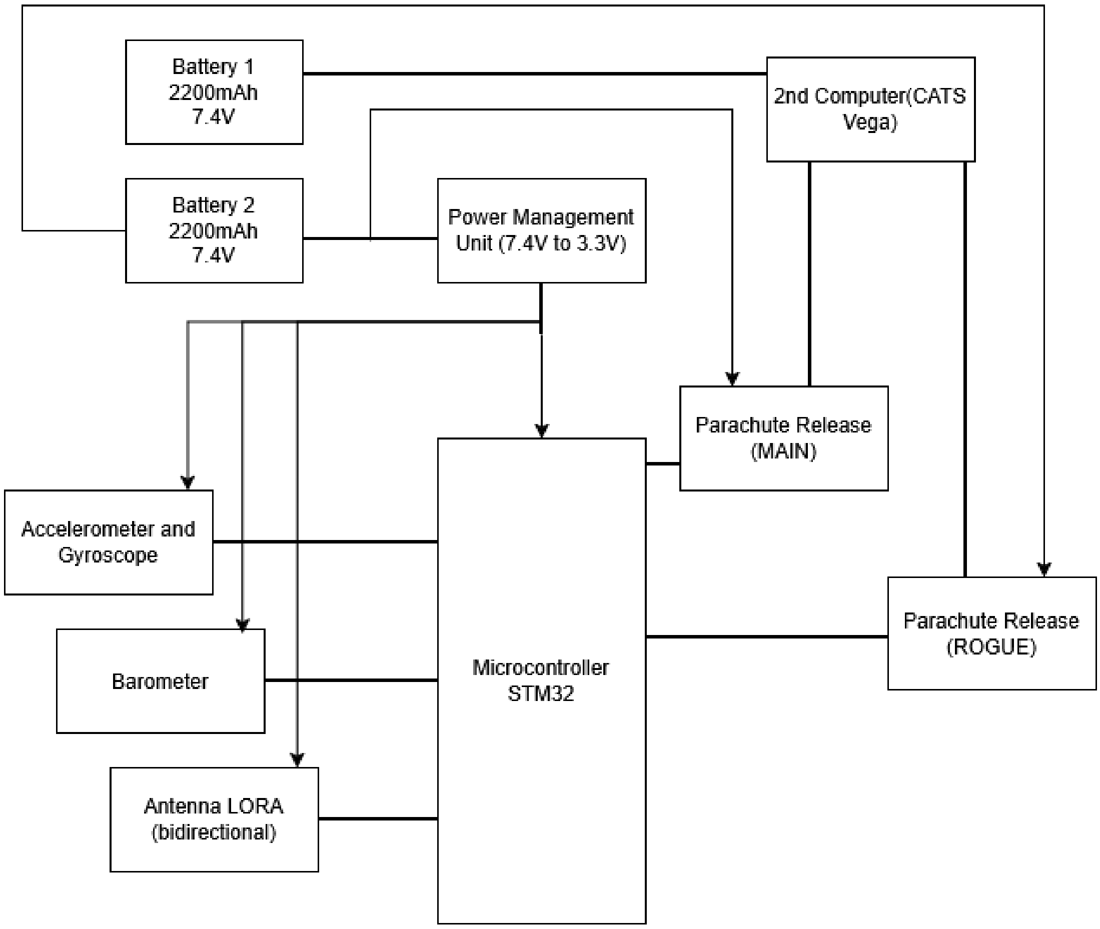


Figure 1: Electrical system diagram

2.1.1 Battery System

The avionics system is powered by Lithium-Polymer (LiPo) batteries. The primary battery used for the SRAD electronics is a Gens Ace 2S1P LiPo battery rated at 7.4 V nominal voltage and 2400 mAh capacity. The battery features a two-cell configuration and a 30C discharge rating, providing sufficient current capability for all avionics loads while maintaining minimal voltage sag. An XT60 connector is used to ensure a reliable and low-resistance power connection.

This battery was selected due to its high energy density, low mass, and robust discharge performance.

A second independent battery powers the COTS flight computer, ensuring redundancy in the event of a failure in the primary avionics power system.

- **Main Battery (SRAD):** 7.4 V, 2400 mAh

- **Redundancy Battery (COTS Flight Computer):** 7.4 V, 2400 mAh

Additionally, external power can be supplied through a pad connector during pre-flight operations, allowing the onboard batteries to remain fully charged until launch.

Parameter	**Value**
Balancer Connector Type	JST-XHR-2P
Brand	Gens Ace
Capacity (mAh)	2400
Voltage (V)	7.4
Length (±5mm)	88
Width (±2mm)	29
Height (±2mm)	17
Net Weight (±20g)	94

Table 1: Specifications of the Gens Ace 7.4V 2400mAh LiPo battery

2.1.2 Power Management Unit (PMU)

The Power Management Unit (PMU) is responsible for regulating and distributing power to the onboard electronics. Instead of using a simple voltage divider, the system employs an LT317A voltage regulator to provide a stable 3.3 V output under varying current loads.

The PMU converts the main battery voltage into regulated 5 V and 3.3 V supply rails required by the avionics systems. For safety purposes, the PMU can be disconnected through an arming pin during handling and integration procedures.

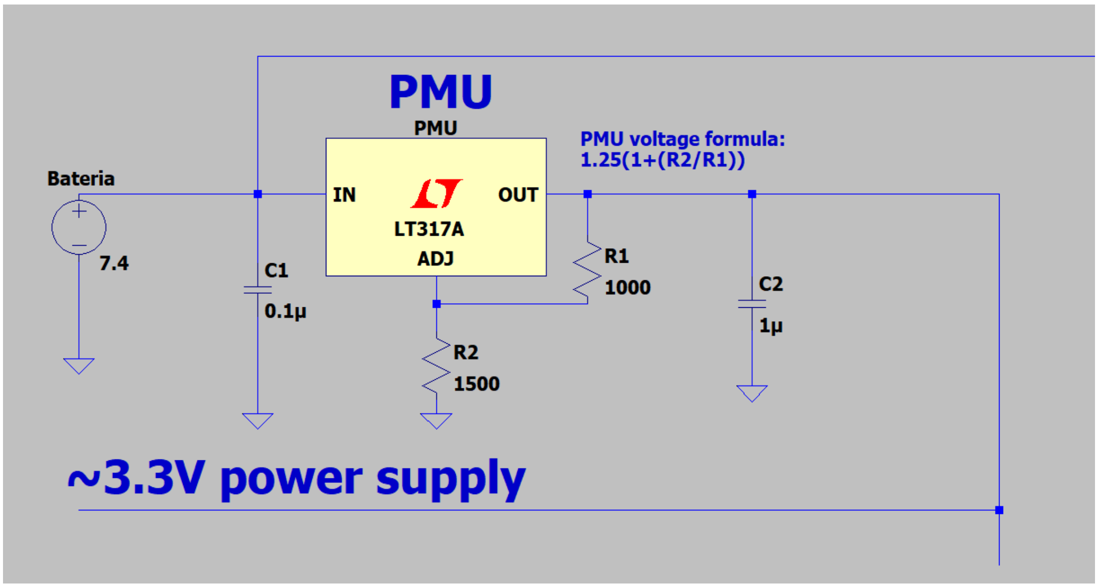


Figure 2: Power Management Unit schematic

2.1.3 Sensors

2.1.4 Parachute Release System

The recovery system consists of two independent parachute deployment mechanisms: a drogue parachute and a main parachute.

The STM32 flight computer controls the deployment sequence using sensor data and onboard flight algorithms. For the main parachute, the STM32 activates a transistor that allows high current from the battery to pass through a low-impedance resistor. The resulting heat ignites a pyrotechnic compound that burns the restraining cable and releases the parachute.

As a redundancy measure, the CATS Vega flight computer can independently trigger deployment through a separate transistor in the event of an STM32 failure.

The drogue parachute is deployed using a servo-actuated latch mechanism. The servo releases the latch, which in turn deploys the drogue parachute and subsequently enables safe deployment of the main parachute.

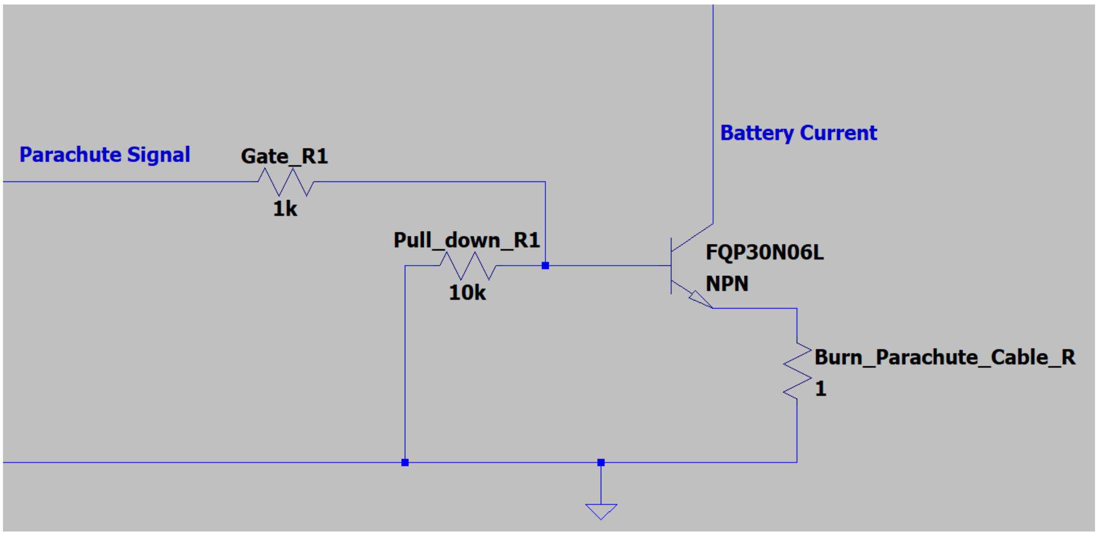


Figure 3: Main parachute electrical diagram

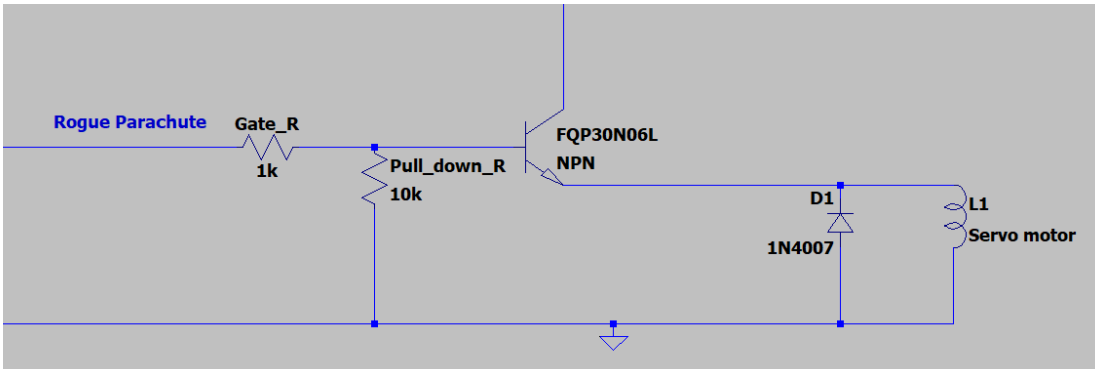


Figure 4: Drogue parachute electrical diagram

3 SRAD Flight Computer Software Architecture

An STM32 microcontroller was selected as the primary flight computer for the SRAD avionics system due to its high reliability, deterministic execution behaviour, and compatibility with the existing commercial off-the-shelf (COTS) avionics used by the team. These characteristics make the STM32 platform particularly well suited for safety-critical aerospace applications, where predictable operation and robust system performance are essential.

A major advantage of the STM32 architecture is its deterministic execution model. Unlike systems that rely on a general-purpose operating system, the STM32 executes firmware either in a bare-metal environment or under a lightweight real-time operating system (RTOS). This eliminates the unpredictability associated with background processes and non-deterministic task scheduling, ensuring that timing-critical operations are executed with consistent latency and repeatable behaviour. Functions such as sensor acquisition, telemetry transmission, state estimation, and recovery system control therefore operate within well-defined timing constraints. This deterministic behaviour significantly improves software validation and testing, which is essential for mission-critical systems.

The STM32 platform also provides hardware commonality with the team's COTS avionics system, which operates as a redundant backup flight computer. Both systems are based on the same STM32 microcontroller family, allowing the same firmware codebase to be deployed across both platforms with minimal modification. This reduces software development complexity, minimises the risk of software divergence between systems, and simplifies debugging, testing, and long-term maintenance.

Firmware development is carried out using **STM32CubeIDE**, the official integrated development environment provided by STMicroelectronics. STM32CubeIDE was selected due to its integrated peripheral configuration tools, native STM32 support, hardware abstraction libraries, and advanced debugging capabilities. Using the official development ecosystem improves maintainability, ensures accurate peripheral configuration, and provides reliable debugging support during both development and ground testing.

The firmware architecture is implemented using **FreeRTOS**, enabling deterministic real-time task scheduling and modular software organisation. Critical subsystems operate concurrently as independent real-time tasks while maintaining predictable execution timing. Separate tasks are allocated for sensor acquisition, telemetry handling, flight-state management, data logging, and recovery system logic, improving scalability, software maintainability, and overall system reliability.

Communication between the SRAD flight computer and external systems is currently implemented using **UART** communication via the RX/TX interface. UART is used for telemetry transmission, debugging, and inter-device communication during development and testing.

Programming and debugging are performed using the **ST-LINK V2** interface, shown in Fig. 5. The ST-LINK V2 provides a stable interface between the development workstation and the STM32 microcontroller, enabling firmware uploading, real-time debugging, memory inspection, and breakpoint analysis.


ST-LINK V2
STM8 & STM32
RST
SWDIO
GND
GND
SWIM
SWCLK
3.3V
3.3V
5.0V
5.0V

The complete hardware setup used during firmware development and testing is shown in Fig. 6.

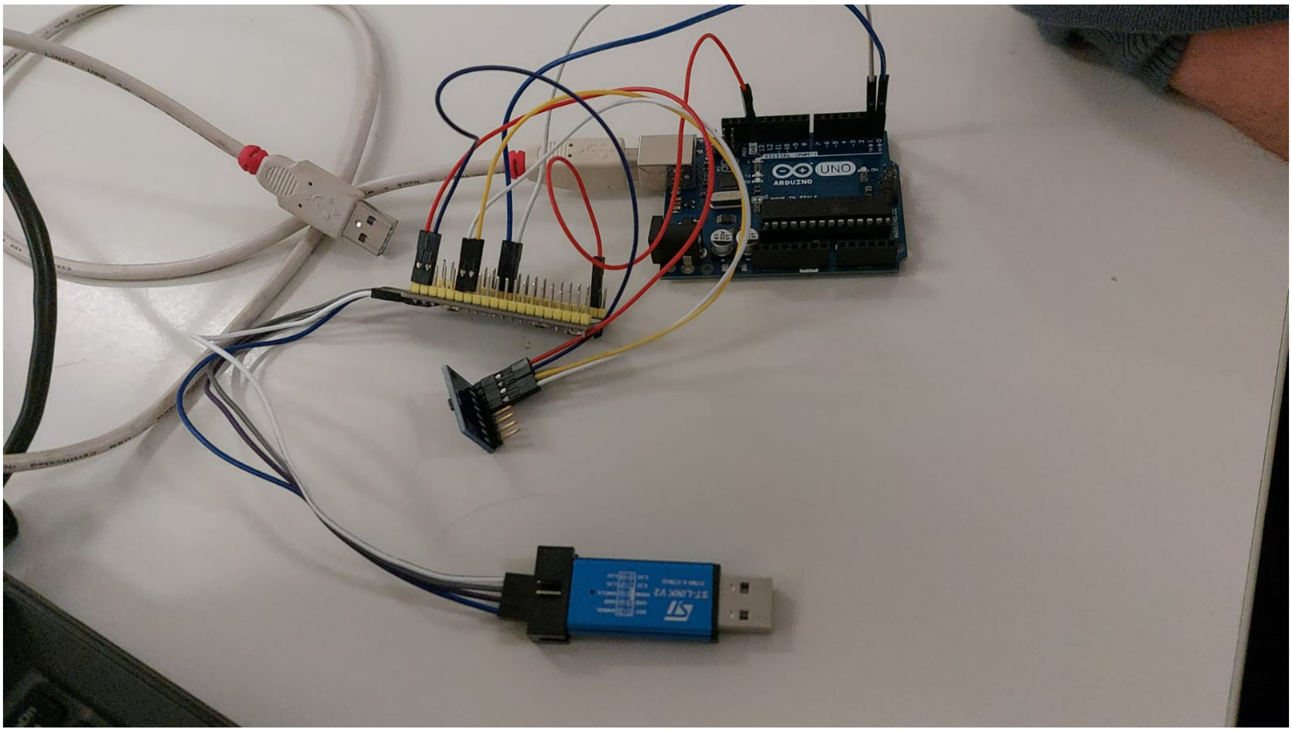
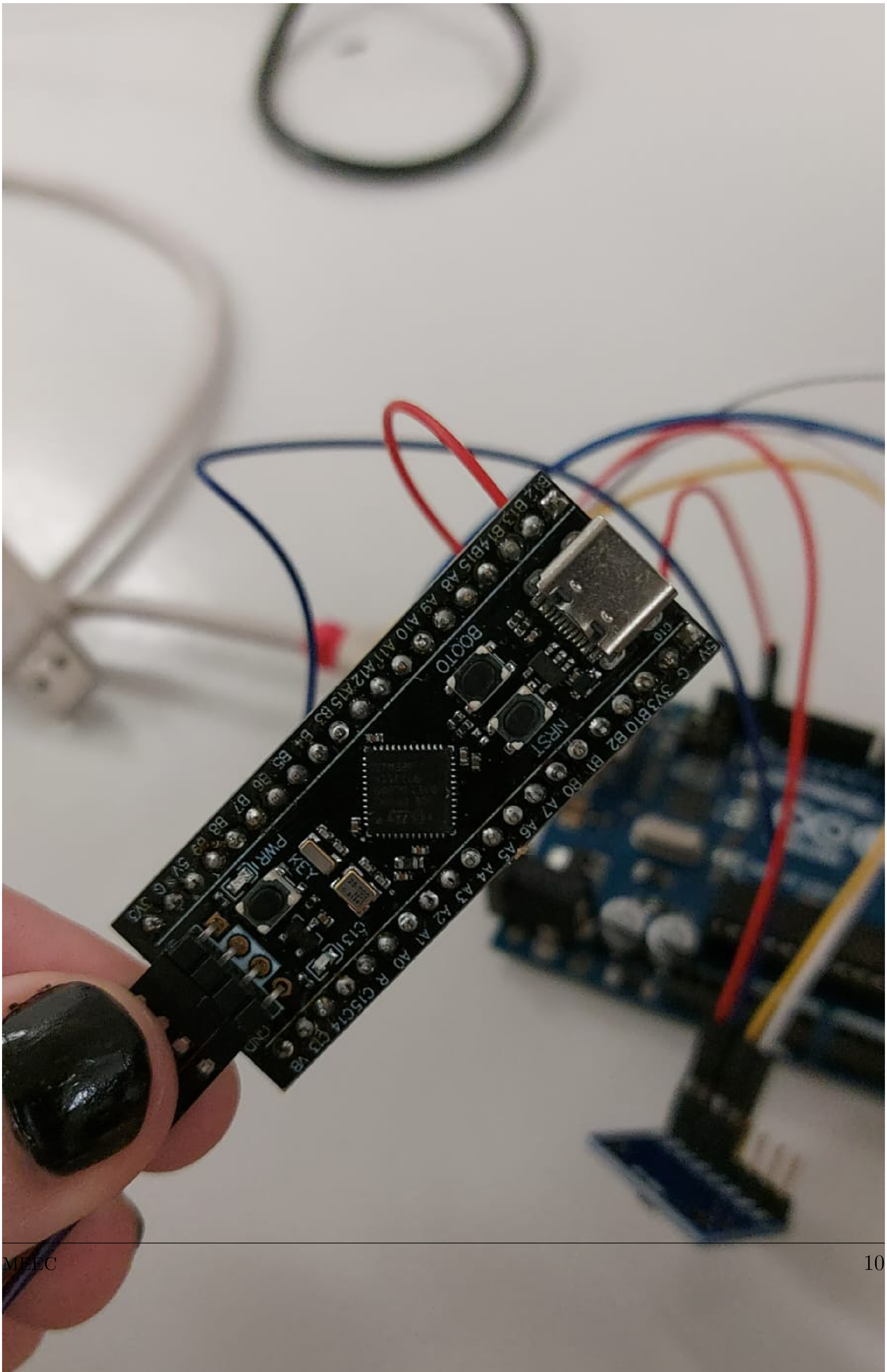


Figure 6: Current SRAD computer development setup

The STM32F411 development board used for firmware implementation is shown in Fig. 7.


BOOT0
NRST
PWR
KEY
GND

For initial communication testing and telemetry verification, UART communication was established between the STM32 and an Arduino Uno development board. In this configuration, inertial measurement unit (IMU) data is transmitted from the STM32 over the UART interface and monitored using the Arduino IDE serial monitor on a ground station computer. This arrangement provides a simple and effective platform for validating serial communication reliability and sensor data transmission prior to integration with the complete avionics stack.

The overall flight architecture incorporates both the SRAD flight computer and the COTS recovery system operating in parallel to provide redundancy and increased operational safety. The COTS recovery system is configured as the primary authority for parachute deployment and will always issue the recovery command when deployment conditions are satisfied. In the event of conflicting commands, the COTS system overrides the SRAD deployment signal. This hierarchical recovery structure introduces an additional safety layer and reduces the likelihood of deployment failure resulting from software or hardware anomalies within the SRAD system.

3.1 Microelectronics

3.1.1 SRAD Electronics

A summary of the primary electronic components used within the SRAD avionics system is provided in Anne, Table 3. The table includes operating voltage ranges, typical current consumption, and estimated power usage where available.

3.1.2 COTS Recovery Electronics

The commercial off-the-shelf (COTS) recovery avionics system used as a redundant backup is also detailed in Annex, Table 4, including its supply requirements and power consumption.

3.2 FreeRTOS

3.2.1 `task_data_acquisition`

Priority: Very High

Description: This task is responsible for continuously reading raw data from the onboard sensors (Barometer, IMU, and GPS) at a strict and fixed frequency (e.g., 100 Hz or 400 Hz). By utilizing the `vTaskDelayUntil` function, it guarantees absolute timing execution without drift. To maintain its high-frequency execution rate and prevent blocking, this task intentionally avoids any complex calculations or data filtering. Its sole responsibility is to efficiently poll the I2C and SPI buses and immediately dispatch the gathered raw data to the relevant queues for downstream processing.

3.2.2 `task_filter`

Priority: High

Description: This task receives the raw data from the sensors and processes it through Kalman filter. Its primary objective is to take noisy sensor readings and calculate clean, accurate estimates for true altitude, velocity, and acceleration. These are the exact state variables required by the flight control state machine to safely and accurately trigger phase transitions.

3.3 Electronics Bay Configuration

The avionics bay provides the internal working volume required to house the SRAD and COTS electronics systems. The available internal dimensions are listed below:

- **Length:** 300 mm
- **Inner Diameter:** 119 mm

The planned PCB arrangement within the avionics bay is illustrated in Fig. 8.

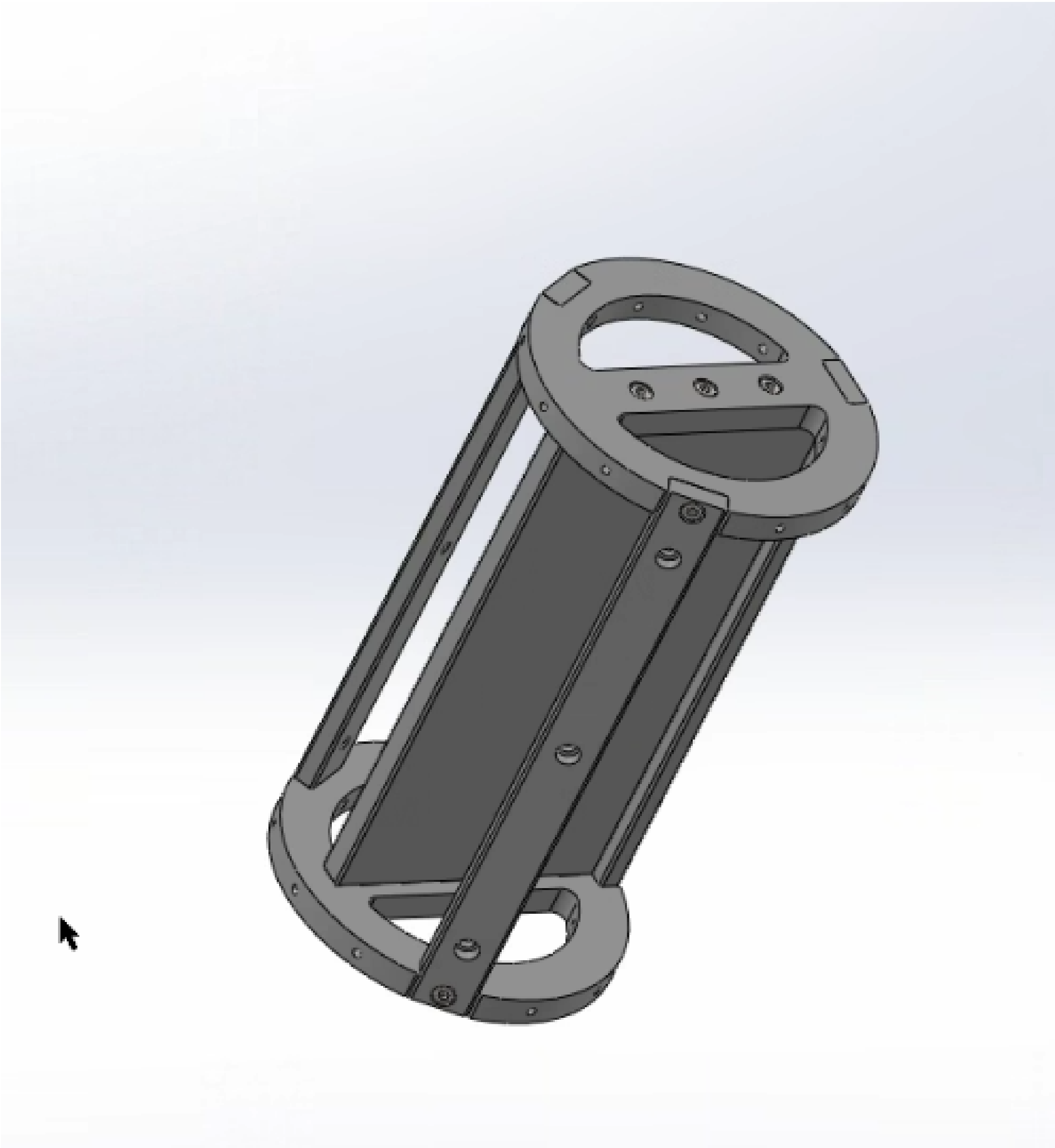


Figure 8: Planned PCB placement inside the avionics bay

Table 2 identifies the function of each PCB within the avionics stack.

Table 2: PCB functional designations

PCB	Designation
PCB 1	SRAD Electronics
PCB 2	COTS Electronics
PCB 3	Power Supply Unit (PSU)

4 Antennas

The primary objective is to establish a reliable telemetry and command link between the rocket and the Ground Control Station (GCS). The system must support real-time telemetry downlink and command uplink throughout the mission profile, from launch to apogee and subsequent descent.

The 433 MHz band offers a superior propagation profile compared to 2.4 GHz, significantly reducing free-space path loss (FSPL). This lower frequency improves signal penetration through potential ground obstacles and atmospheric phenomena, which is critical for maintaining link margin at long ranges with low transmitter power.

The primary justification for selecting 433 MHz is the minimization of FSPL. The theoretical signal loss at apogee (3.2 km) is calculated using the Friis Transmission Equation in logarithmic form below. Considering d = 3.2 km (Maximum range) and f = 433.15 MHz the loss is approximately 95.2 dB, the 32.44 factor comes from the conversion of distance to km and Hz to MHz.

The link margin will serves as a measure of reliability, representing the safety buffer between the received signal and the system's sensitivity floor.

- Tx Power (LoRa Ra-02): $+20\,\text{dBm}$
- Tx Antenna (Rocket Omni): $0\,\text{dBi}$
- Path Loss (at 3.2 km): $-95.2\,\text{dB}$
- Rx Antenna (GCS Yagi): $+9.7\,\text{dBi}$
- Cable/Connector Losses: $-2\,\text{dB}$

The estimated Received Power is $P_{rx} = 20 - 95.2 + 9.7 - 2 = -67.5$ dBm. Given the LoRa SX1278 sensitivity of approximately -130 dBm (at SF12), the final margin is Margin $=$ $M\arg in = P_{rx} - P_{sensitivity} = -67.5 - (-130) = 62.5\text{dB}$. A 62.5 dB margin is highly robust, allowing the system to tolerate significant tumbling (polarization mismatch), Yagi misalignment, or interference while maintaining a signal lock.

The physical dimensions of the Yagi elements (Reflector length approx. 329 mm) directly correlate to the 433 MHz band. Using the standard approximation for a half-wave element $\lambda = \frac{c}{f} \approx \frac{300}{433} \approx 0.693m \Rightarrow \lambda/2 \approx 346mm.$

The measured element lengths (e.g., Reflector at 329mm) are shorter than the standard free-space half-wave dipole (346 mm) to account for the Boom Correction Factor. Since the boom is made of conductive aluminum, it electrically interacts with the parasitic elements. When

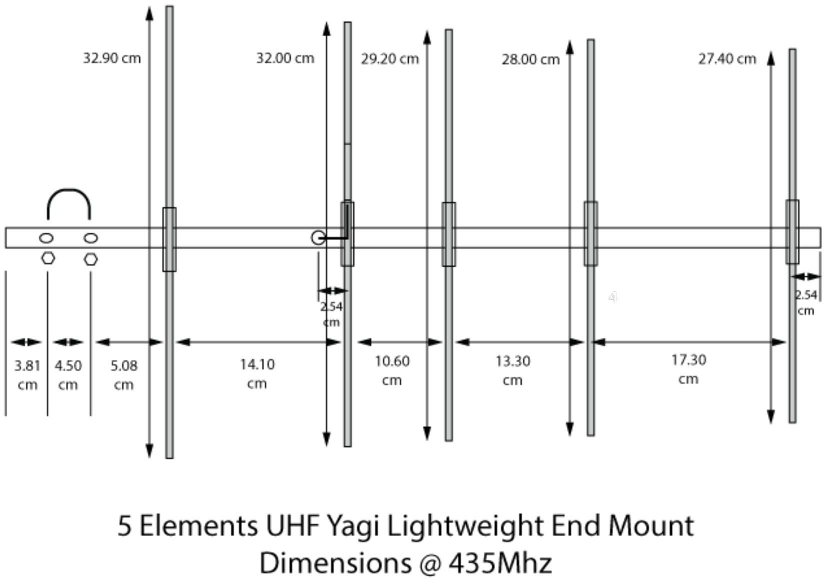


Figure 9: 5-element Yagi-Uda antenna geometry

an element passes through or connects to a metal boom, the boom acts as a shorted turn, effectively lengthening the element electrically. To compensate for this and maintain resonance at 433.15 MHz, the physical elements must be shortened. For a 20 mm boom, elements are typically reduced by 15-20 mm compared to a non-conductive boom:

$\lambda/2 = 346\text{mm} - 17\text{mm} \approx 329\text{mm}$

The choice of a Gamma Match feed is also dictated by the conductive aluminum boom. In this design, the center of the driven element is electrically grounded to the boom. This creates a strong mechanical bond and eliminates static charge buildup. A standard Yagi driven element has a low natural impedance (20 Ω). The Gamma match transforms this low impedance up to the 50 Ω required by the coaxial cable and transceiver. The slider mechanism on the Gamma match allows for precise VSWR tuning at 433.15 MHz, compensating for any environmental detuning at the launch site.

The electromagnetic performance of the 5-element Yagi-Uda array was analyzed using MATLAB's Antenna Toolbox. The simulation model uses the exact physical dimensions defined in the hardware design.

Distinct from the real antenna, this simulation models a direct split-feed dipole excitation rather than the Gamma Match mechanism or the conductive boom. Consequently, the simulation results present the resonance of the shortened elements (approximately 440 MHz) prior to the inductive correction provided by the Gamma Match.

The simulated 3D radiation pattern shown in Figure 10 confirms that the parasitic arrangement successfully focuses RF energy into a single dominant forward lobe. The analysis yields a maximum directivity of 9.41 dBi, which provides a signal amplification of approximately 8.7

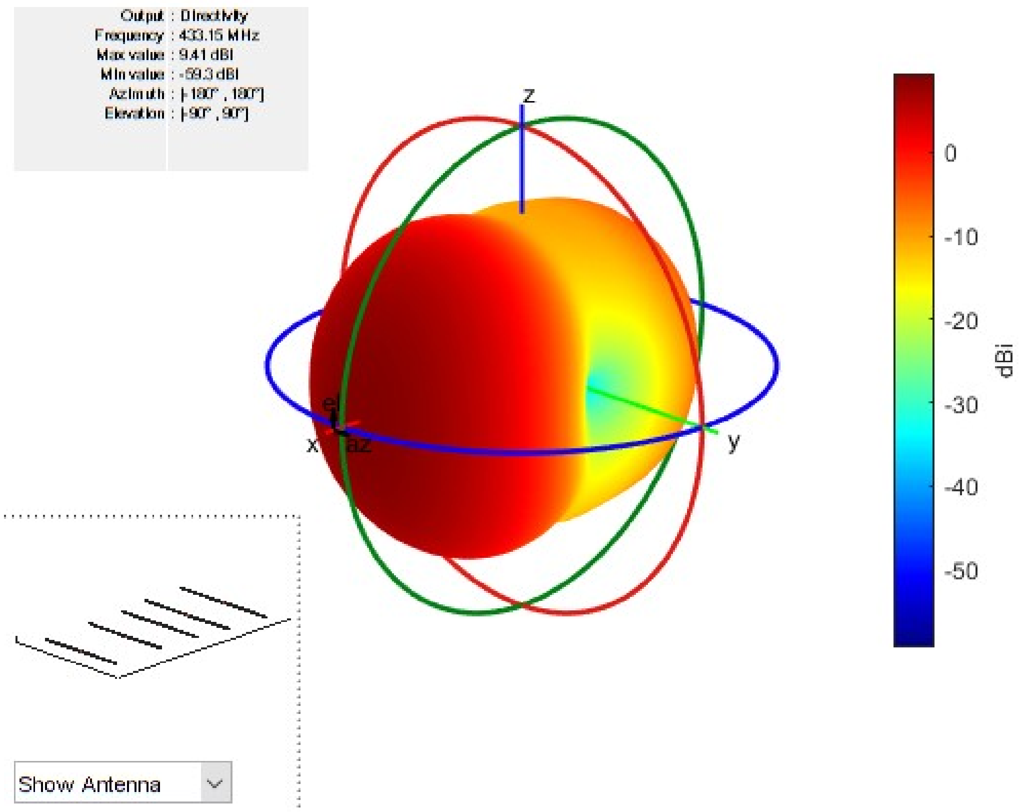


Figure 10: Simulated 3D radiation pattern of the 5-element Yagi-Uda array

times compared to an isotropic radiator. The beam is strictly aligned with the boom axis (Azimuth 0°, Elevation 0°), validating the element spacing and phasing.

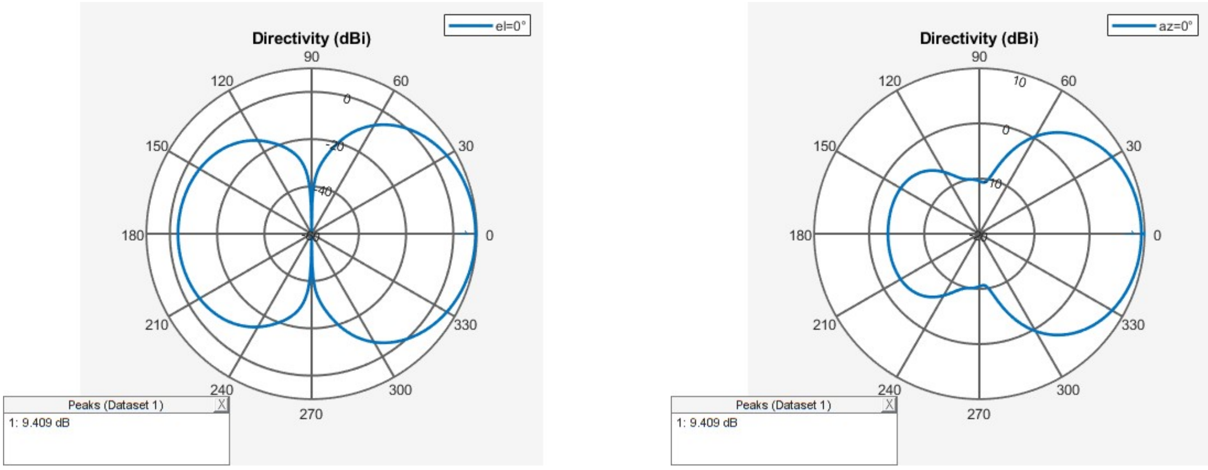
To define the operational requirements for the Ground Station, we analyze the 2D pattern cuts. The vertical cut shown in Figure 11b displays the pattern perpendicular to the elements. The main beam is broad, offering a Half-Power Beamwidth (HPBW) of approximately 60º. The horizontal cut shown in Figure 11a displays the pattern parallel to the elements. The pattern exhibits deep nulls at 90º, which aids in rejecting interference from lateral sources.

The broad HPBW implies that the ground station operator has a generous tracking window. The antenna can be misaligned by up to 30º from the rocket's position while still maintaining a signal strength within 3 dB of the peak. This eliminates the need for high-precision automated tracking systems.

A critical aspect of this design is the interaction between the physical element lengths and the feed mechanism. Figure 12 shows the return loss (S_{11}) of the simulated structure.

The simulation shows the resonance dip centered at approximately 440 MHz, rather than the target 433.15 MHz. This is an expected result because of the boom correction effect and the gamma match feed role.

The physical elements were intentionally shortened (e.g., Reflector 329mm instead of the theoretical 346mm) to compensate for the conductive aluminum boom, which electrically lengthens elements in the real world. Since the MATLAB simulation models free-space elements (no boom), it accurately reports that these short elements resonate at a higher frequency (440 MHz).



(a) E-plane horizontal cut

(b) H-plane vertical cut

Figure 11: Radiation pattern plane cuts

In the physical assembly, the Gamma Match provides the necessary inductance to lower the resonant frequency from 440 MHz to 433.15 MHz, while simultaneously transforming the feed impedance to 50Ω.

This simulation confirms that the physical dimensions are correct for a boom-mounted, Gamma-matched implementation. If the simulation had shown resonance at 433 MHz using these short lengths without the boom model, the physical antenna would likely have been tuned too low (e.g., 420 MHz) once built.

The analysis confirms that the 5-element Yagi provides the necessary 9.4 dBi gain to close the link budget at the 3.2 km apogee. The combination of high forward gain and a manageable 60º beamwidth satisfies the mission requirement for a reliable, manually trackable uplink/-downlink.

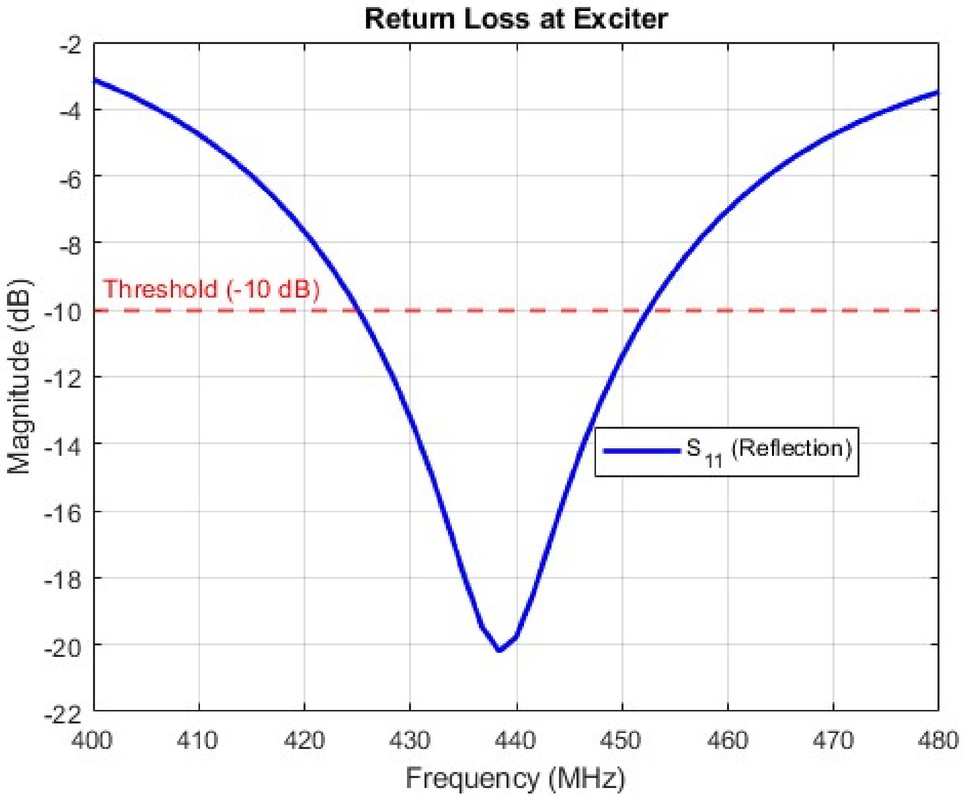


Figure 12: Simulated return loss (S_{11}) of the antenna

5 Ground Station

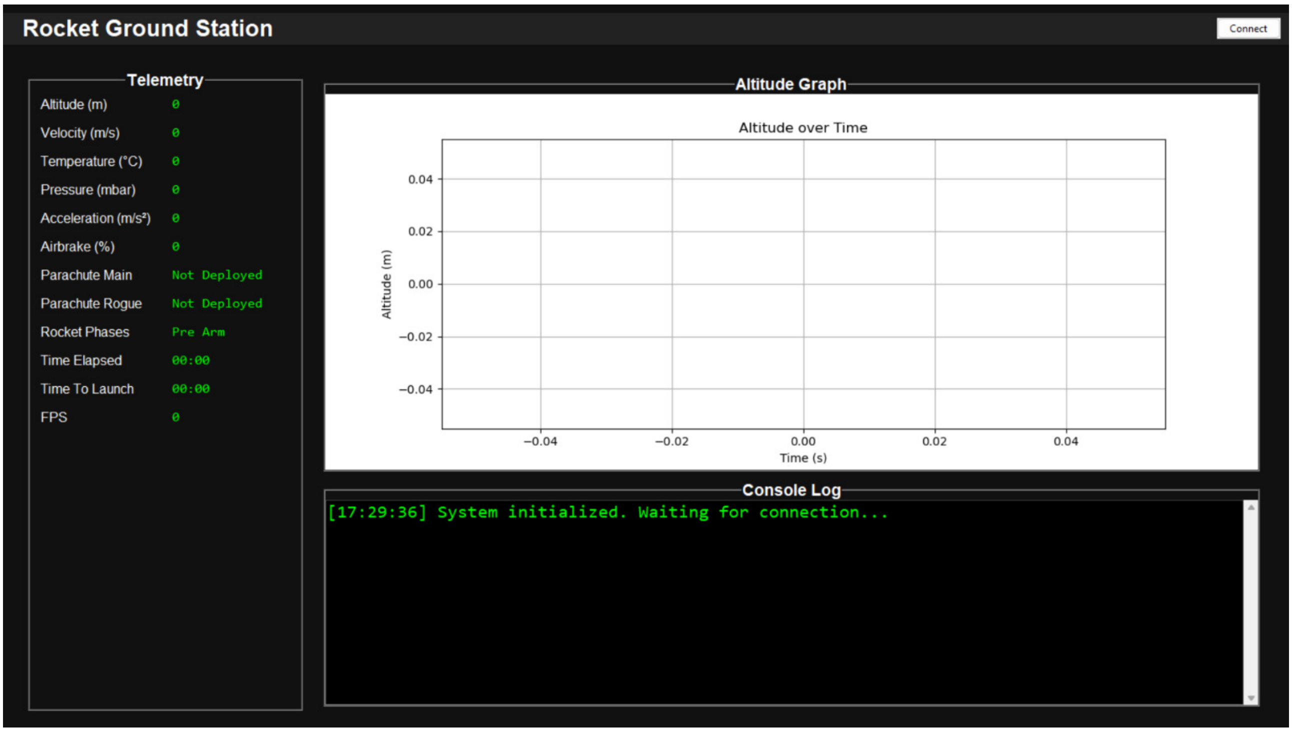


Figure 13: Current Ground Station

TODO:

- Add simulation CSV data input
- Display position in 3D

6 Control System / Dynamics

6.1 Active control

The active control system of the *ASTRO Rocket* is exclusively dedicated to the deployment of airbrakes, which are used to increase aerodynamic drag and reduce velocity in order to achieve the target apogee. Other forms of active control, such as fin actuation, are prohibited by the European Rocketry Challenge (EuRoC) regulations.

The airbrake control is implemented using PID controllers running on one of the onboard microcontrollers. This controller communicates with the servo motor system, which actively deploys the airbrakes by rotating a geared mechanism. The PID controller is designed based on the relationship between the *drag coefficient* and the *Mach number* for each airbrake deployment level. Velocity measurements are used as the primary feedback variable to determine the appropriate control action required to reach the target velocity at each altitude. The altitude is estimated by correlating the pressure measured by the onboard barometer with the International Standard Atmosphere (ISA) pressure model. Further details regarding the experimental implementation and results will be presented in future revisions of this document.

6.2 Sensing System

The sensing and control subsystem features a more sophisticated architecture. All sensors are connected to an onboard microcontroller dedicated to data handling and signal processing. The acquired data are processed through a Multiplicative Extended Kalman Filter (MEKF) to enhance state estimation accuracy and suppress measurement noise. The MEKF is a well-established algorithm widely used in spacecraft attitude determination systems due to its high precision [?].

In the context of apogee estimation, the MEKF contributes by providing accurate attitude quaternion estimates derived from the magnetometer, gyroscope, and accelerometer measurements. When the pitch angle, inferred from the estimated quaternion, approaches zero, the rocket is considered to have reached the apex of its parabolic trajectory, indicating the apogee point.

7 Simulations

The rocket simulation was performed using *RocketPy*, a well-established and thoroughly tested open-source Python library for trajectory simulation and analysis of sounding rockets and high-power rockets. The latest stable version (v1.1.1) was used for this project.

7.1 RocketPy Framework Structure

RocketPy follows a modular architecture that separates the simulation into four primary components:

- **Environment Module:** Defines atmospheric conditions such as temperature, pressure, wind profiles, and gravity models. The module supports multiple atmospheric datasets, including the Standard Atmosphere model, custom atmospheric inputs, and weather reanalysis or forecast models (e.g., ECMWF, GFS).

  For this project, historical weather data from the EUROC competition was used to enable realistic retrospective flight simulations. Atmospheric data can be obtained from sources such as the Copernicus Climate Data Store: `https://cds.climate.copernicus.eu/`

- **Motor Module:** Describes the propulsion system using thrust curves, grain geometry, propellant properties, and nozzle characteristics. RocketPy supports solid, liquid, and hybrid motor configurations, allowing detailed modelling of mass flow and centre of mass variation during burn.

  In this work, a solid rocket motor is used.

- **Rocket Module:** Defines the complete vehicle geometry and mass properties, including aerodynamic surfaces (nose cone, fins, and tail sections), inertial properties, drag coefficients (powered and unpowered flight), and recovery systems such as parachutes.

- **Flight Module:** Integrates all previous components to simulate the full trajectory using a six degrees of freedom (6-DOF) numerical solver. The simulation accounts for aerodynamic drag, gravity, wind effects, thrust, rail guidance, and parachute deployment events.

7.2 JSON-Based Configuration Architecture

To improve workflow efficiency and enable rapid iteration during the design process, a custom JSON-based configuration system was developed. This approach offers several advantages over hard-coded Python scripts:

- **Centralized configuration management:** All rocket parameters are stored in a single `rocket.json` file. This allows non-expert users to modify simulation parameters without interacting directly with the source code.

- **Version control:** Design iterations are tracked through JSON file versioning, enabling a clear history of configuration changes.

A custom loader function, `load_flight_from_json()`, parses the JSON configuration and instantiates the corresponding RocketPy objects. The JSON structure mirrors the RocketPy architecture:

```
{
  "path": ".\\specifications\\",
  "environment": { ... },
```

```
  "motor": { ... },
  "rocket": { ... },
  "flight": { ... }
}
```

Aerodynamic Modelling

The rocket uses separate drag coefficient curves for powered and unpowered flight phases, accounting for changes in base drag during motor burn. These curves can be generated using aerodynamic analysis tools such as *RASAero*.

7.3 Air Brakes System (Currently Disabled)

The configuration includes provisions for an active air brakes system located at 1.30 m along the rocket body. However, this system is currently disabled in the simulation.

When enabled, the air brakes system includes:

- Dynamic drag coefficient variation as a function of deployment level and Mach number
- A control algorithm operating at 10 Hz sampling rate
- Closed-loop integration with flight state for apogee targeting

This system is intended for future iterations to enable precise altitude control and achieve competition targets of approximately 3000 m. However, for the current design, the system was removed from the physical rocket to improve simplicity and reliability, as this is the team's first rocket design.

7.4 Redundancy Checking

To validate simulation accuracy, *OpenRocket* is used as an independent verification tool. Simulation results are considered acceptable if the following parameters agree within 5%:

- Apogee altitude
- Maximum velocity
- Parachute deployment timing

7.5 Document Management

A RocketPy simulation requires structured input data defining the environment, motor, and rocket configuration.

Environment Definition

The `Environment` class stores atmospheric and geographic data required for trajectory prediction. It is defined using:

- Latitude
- Longitude
- Elevation
- Atmospheric model specification (year, month, day, and UTC time)

Motor Data Format

RocketPy supports solid, hybrid, and liquid propulsion systems. Since a solid motor is used, thrust curve data is required.

The thrust curve file is structured as follows:

- Header line: common name, diameter, length, propellant weight, total weight, manufacturer
- Subsequent lines: time (s) and thrust (N)

Rocket Definition

A Rocket object is defined after the motor and includes aerodynamic surfaces, recovery systems, and rail guides. Key parameters include mass properties, aerodynamic coefficients, and geometric configuration.

Table 3: SRAD avionics electronic components quick specifications

Device	Supply (V)	Current	Power (mW)	Interface	Role	Notes
STM32F411 Microcontroller	1.8–3.6	–	–	I2C / SPI / UART	Main flight computer	Cortex-M4 MCU
ICM-45686 6DOF IMU	3.3	34 mA	112	I2C / SPI	Inertial sensing	High-rate motion tracking
ADXL375 3DOF Accelerometer	2.0–3.6	145 µA	0.5	SPI / I2C	High-G acceleration	Up to ±200 g
LIS2MDL Magnetometer	1.71–3.6	200 µA	0.66	I2C	Heading reference	Low-power compass
MS5607 Barometer	1.8–3.6	1.4 mA	4.6	I2C / SPI	Altitude estimation	High-resolution pressure sensor
NEO-M9N GPS Module	3.6	–	–	UART	Positioning	Multi-constellation GNSS
SX1278 LoRa (Ra-02)	1.8–3.3	105 mA	346	SPI	Telemetry link	Long-range RF comms
SparkFun microSD Module	3.3	20 mA	66	SPI	Data logging	Flash storage interface

Table 4: COTS avionics system specifications

Device	Supply (V)	Current	Power (mW)	Interface	Role	Notes
CATS Vega Flight Computer	7–24	100 mA	321.75	UART / CAN (var.)	Recovery control system	Integrated flight computer + deployment logic

Table 5: Bill of Materials

Item	**Status**	**Brand**	**Supplier**	**Qty**	**Unit**	**Total**
STM32F411 Development Board	In Lab	STMicroelectronics	Fruugo	1	8,95	8,95
Cats Vega	Need to order	catsystems	catsystems.io	2	395,73	791,46
GNSS M6N	In Lab	u-blox	Donated	2	0	0
LSM6DSMTR (Accel + Gyro)	Need to order	STMicroelectronics	Farnell / DigiKey	1	3,07	3,07
MS5607-02BA Barometer	Need to order	Measurement Specialties	DigiKey	1	2,8	2,8
LIS3MDL Magnetometer	Need to order	STMicroelectronics	PTRobotics	1	9,16	9,16
LiPo Battery 2200mAh 7.4V 2S1P	Need to order	Gens Ace	Techinn	2	17,95	35,9
LT317A PMU	Shipping	–	Mauser PT	2	0	0
Capacitor 1uF	In Lab	–	Mauser PT	2	0	0
Capacitor 0.1uF	In Lab	–	Mauser PT	2	0	0
FQP30N06L MOSFET	Shipping	–	Mixtronica	4	0	0
Antennas	Manufacturing in house	Nuar-Astro	–	2	20	40
LoRa SX1278 RA-02	Shipping	Ai-Thinker	LojaPM	2	9,89	19,78
				Grand Total		**911,12**